

EG-M36 Assignment Cover Page – 2024/25



Swansea University
Prifysgol Abertawe

Group Name	GROUP - F
Submission date	4 th April 2025
What parts of this assignment are you most pleased with?	QFD Chart, calculating the safety margin and performing risk assessment, HAZOPS, Identifying the reliability requirements. Calculating the system reliability and performing Fault Tree Analysis.
Were there any areas you found challenging? If so, please specify	Drawing Load (stress) – Strength graph, Benchmarks because it was hard to find the values for the materials that I was looking for.
Are there elements you'd like feedback on? If so, please specify	Improved CAD, Improved failure reliability calculations

Task Sheet 1



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Role/Responsibility Matrix (in the case of 5 students)

No.	Student Name	Student ID	Task Sheet No.	Role/Responsibility (person assigned with role takes the lead)	Skills needed
1	Varghese Mathew	2456476	2b, 2c, 3	- Identifying environmental/usage conditions - Identifying reliability requirements and goal - Benchmarking	- Researching - Data collection/analysis - Calculation - Systematic analysis
2	Praveen Kumar Mohandas	2380424	3, 4	- Benchmarking - QFD	- Researching - Systematic assessments (quantitative/qualitative)
3	Arsenios Konstantinou	2101048	7a, 7b	- Block Diagram Analysis (BDA) - Fault Tree Analysis (FTA)	- Data analysis - Calculation - Systematic assessment
4	Shreyas Ramesh Shetty	2502923	6, 8	- HAZOPS - FMEA	- Systematic assessments (quantitative/qualitative) - Calculation & analysis
5	Akshay Ravindra Taware	2428966	5, 9a, 9b	- Programme risk assessment - Load (stress) – Strength Analysis	- Systematic assessment - Calculation & analysis
6	NA (Group task)	NA (Group task)	2a, 10	- Failure Classification, degree and causes of failures of the current design - Design (propose at least two improved designs)	- Critical evaluation - Material selection - Conceptualisation - Sketch, drafting and design

Task Sheet 2a

Failure classification, degree and causes of failures



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No.*	Name of Unit(s)	Failure classification (system/component)	Degree of failure (complete/partial)	Causes of failure	Remarks (e.g. references)
1	Rotor Disc (AMMC)	Material loss over time Chances of over heating	Partial :Small crack can result in reduced brake efficiency	Defects in material, deformation is an issue.	Adebisi AA, Maleque MA, Rahman MM. Metal Matrix Composite Brake Rotor: Historical development andproduct life cycle analysis.
2	Rotor Disc	Can cause fatigue Surface degradation might happen	Complete: brake loss will happen	Instant temperature changes Non-uniform heat dissipation	International Journal of Automotive and Mechanical Engineering [Internet]. 2011 Dec 30;4:471–80. Available from: https://doi.org/10.15282/ijame.4.2011.8.0038
3	Brake pad	Brake fade Corrosion is an issue	Partial: due to low friction Complete: can make way to damage the rotor	Grade of material Harsh driving is a reason	Brake Pad Failure Modes and their Consequences – Part 2 [Internet]. Engineering Bulletin. 2018 Apr p. 2–8. Available from: https://www.gbscouncil.com/wp-content/uploads/2018/04/BULLETIN-No.02.04.18.pdf
4	Shim (Rubber coated)	Rust take place in metal layer Due to Continuous stress	Partial: there will be some noise Vibrations will be there	Above 300 degree rubber gets weakened.	Khuntiptong A, Tangchaichit K, Polnumtiang S, Ladbut N, Wansasueb K. Study of brake pad
5	Shim (Rubber coated)	Hardening of rubber layer. Metal deformation	Complete: total loss of damping property.	Shims can get loss its property when it's affected by corrosion	Shim modification to improve stability against high frequency squeal noise by finite element analysis. Engineering Journal [Internet]. 2022 Sep 30;26(9):25–34. Available from: https://doi.org/10.4186/ej.2022.26.9.25

Task sheet 2a

Failure classification, degree and causes of failures



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No.*	Name of Unit(s)	Failure classification (system/component)	Degree of failure (complete/partial)	Causes of failure	Remarks (e.g. references)
6	Calliper (Al)	Stress and fatigue Overheating is a major issue	Due to heavy load Low maintenance.	Due to heavy load Low maintenance.	Keshavaram BN. Aluminum alloys for automotive disc brake calipers. SAE Technical Papers on CD- ROM/SAE Technical Paper Series [Internet]. 1999 Mar 1; Available from: https://doi.org/10.4271/1999-01-0346
7	Calliper (Al)	Integrity of material get lost	Complete: there will be high risk of hydraulic brake failure	Exposure to harsh environmental condition	
8	Brake fluid	Chemical degradation	Partial: boiling of the fuel Complete: vapor lock	Not doing replacement of fluid in correct time lead to function	Kao M j, Tien D c, Ting C c, Tsung T t. Hydrophilic characterization of automotive brake fluid. Journal of Testing and Evaluation [Internet]. 2006 Mar 17;34(5):400–4. Available from: https://doi.org/10.1520/jte14254

Task Sheet 2b

Identifying environmental/usage conditions

*If necessary, add/delete the number of rows and pages for references



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No.*	Name of Unit(s)	Environmental condition	Usage condition
1	Rotor Disc (AMMC)	Can withstand thermal conditions without cracking Works well in a cold climate, corrosion resistant Low dust generation, 113MPa, thermal expansion 3.8	Excellent in hypercars, Good performance in moderate speed braking Good for urban driving
2	Brake pad (Metallic)	Operating range: 50°C-600°C, good heat dissipation, Need coating to resist corrosion in humid condition.	Recommended for steep down hill braking, Required to monitor rotor wear as pads are abrasive.
3	Brake pad	Need coating to resist corrosion in humid condition.	Great for long distance , Produce dust and noise Performs well in city driving
4	Shim	Operating temperature range: 30°C to 400°C, Minimum impact in dust condition, corrosion resistance	It is best for passenger cars and light duty SUV In highway it give good performance but need proper ventilation.
5	Shim	Brake noise may increase if moisture gets trapped between the shim and brake pad. Functions well in rain	Not proposed for continues heavy braking
6	Calliper (Al)	Good thermal conductivity, High temperature vulnerability Aluminium has excellent natural corrosion resistance	For normal driving it performs well. In heavy braking challenges like heat, structural integrity
7	Brake fluid	Boiling point is around 205°C , corrosion risk Absorb moist from air over time	Continuous heavy braking not suggested
8	Brake fluid	Sludge and deposit formation	Provide consistent performance in normal Compatible with ABS but slower in extreme condition

Task Sheet 2c

Identifying reliability requirements and goal



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No.*	Name of Unit(s)	Reliability requirements (e.g. in terms of design, installation, maintenance, operation and test)	Other requirements (e.g. performance, size, shape, mass, cost, manufacturing, standards, government regulations, IP requirements, sustainability, etc.)	Goal (% reliability improvement)
1	Rotor disc	Fatigue & impact testing ASTM E6060 Optimised ventilation channels	Hot isostatic pressing treatment improve properties. Laser surface texturing to improve fatigue life.	26.5-31mm thickness can improve to 40%
2	Rotor Disc	Drilled rotor reduces weight Increase rotor thickness by 10-15%	ISO 9001, ECE R90, and ASTM B595 is cost effective Stress-optimized calliper shape (ribs, hollow sections)	40%
3	Brake pad	Use high advanced alloy Optimised curved geometry	Advanced forging technique, improve precision SAE 661- focus on friction performance testing	40%
4	Brake pad	Micro structured surface Proper bedding-in procedure	Use hybrid alloys to reduce cost	40%
5	shim	Use round edge to prevent wear Use predictive maintenance	Implement automated quality control system Cost can raise around 10-30%	40%
6	shim	Use reinforced ribs	Standard like ISO, SAE, ECE help to meet reliability condition	40%
7	calliper	Detailed air flow path Slotted vent channel in body	Use ultrasonic testing to detect issues, improve casting IATF 16949: ensure high quality calliper	40%
8	Brake fluid	Minimise moisture contamination Hermetically sealed packaging	Use advanced corrosion inhibitors will neutralise acids Maintain proper viscosity at extreme temperature	40%

Task Sheet 3

Benchmarking

*If necessary, add/delete the number of rows and/or columns, and pages for references

Company/Competitor* Performance Characteristics*	'Our Company'	Competitor 1 (TRW)	Competitor 2 (MTEC)	Competitor 3 (Brembo SGL)	Competitor 4 (Bugatti)	Competitor 5 (Surface Transforms)
Material Type	6092/SiC/25p-T6 (Al MMC)	Grey Cast Iron	High-Carbon Steel	Carbon-Silicon Carbide (C/SiC)	Titanium Alloy (Ti-6Al-4V)	Ceramic Matrix Composite
Density (Kg/m³)	2840 (1)	7100-7300 (4)	7500-7850 (7)	2000 (10)	4400-4500(15)	1970 (18)
Thermal Conductivity (W/m·K)	180-200 (2)	40-50 (5)	19-52 (8)	101.33 (13)	21.6 (15)	16-25 (19)
Compressive Strength (MPa)	300-500 (1)	500-650 (4)	400-650 (8)	112 (11)	>800(15)	250-400 (21)
Max Operating Temp. (°C)	340-400 (1)	550-600 (4)	427 (8)	663.7 (12)	350 (16)	1200-1600 (20)
Coefficient of Friction (μ)	0.33 - 0.4 (3)	0.35-0.55 (6)	0.11 - 0.14 (9)	0.4-0.52 (14)	0.35 - 0.50 (17)	0.3 - 0.40
Wear Resistance	Moderate - High	Moderate	High	Very High	High	Extreme
Brake Pad Life (vs. Cast Iron)	0.8x-1.2x	1.0x (Baseline)	1.2 x - 1.5 x	2.0 x- 3.0x	1.3x - 1.8x	3.0 x – 5.0 x
Cost	Medium – High	Low	Medium	Very High	Extremely High	Extremely High

References For Task Sheet 3



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References For Task Sheet 3



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Task Sheet 4

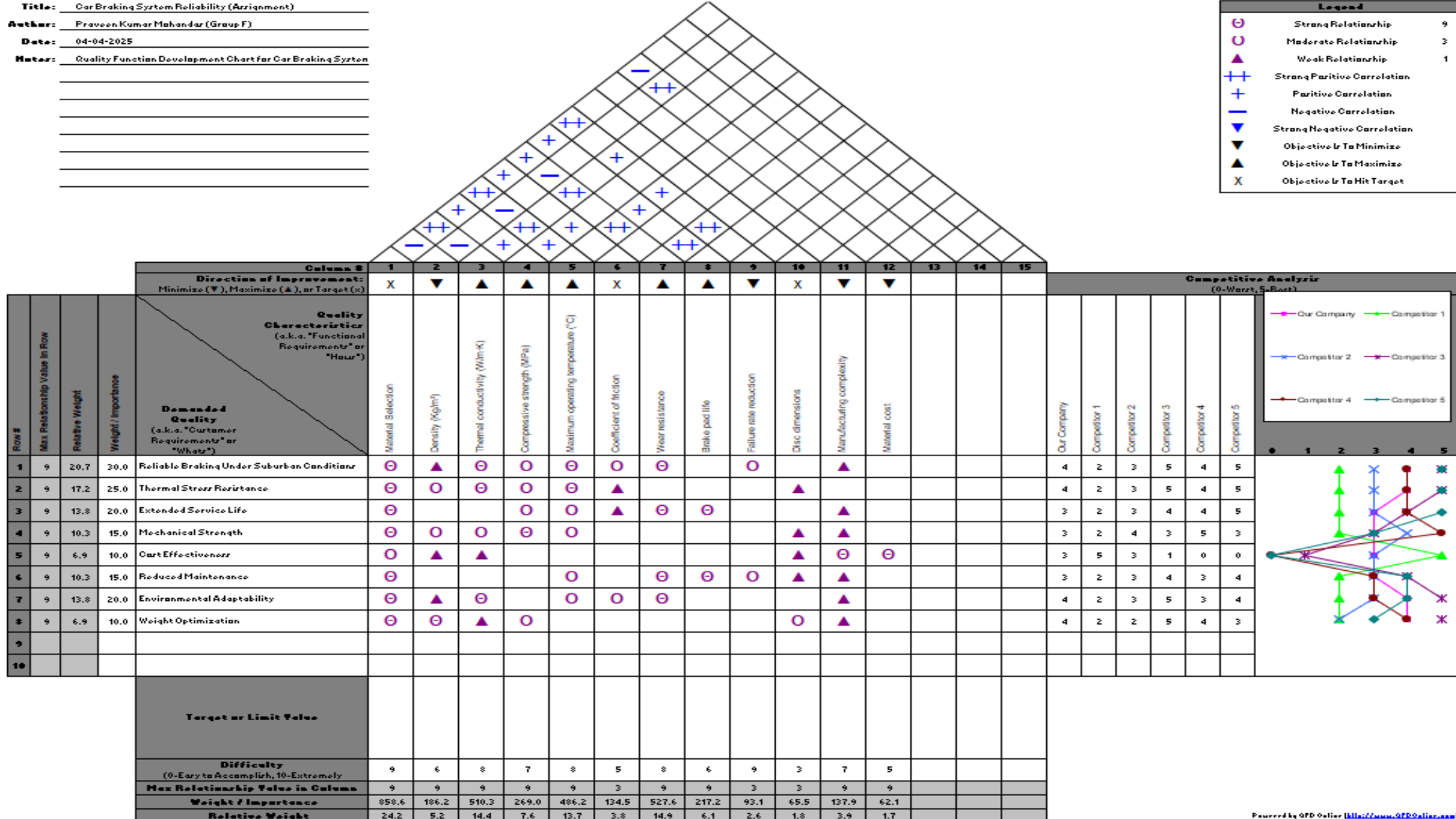
QFD

(Use the Excel template to complete, only one QFD chart is required)



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Title: Car Braking System Reliability (Assignment)
Author: Praveen Kumar Mahendar (Group F)
Date: 04-04-2025
Notes: Quality Function Development Chart for Car Braking System



Task Sheet 5

Programme Risk Assessment

No.	Question	Answer (Yes/No)	Degree of Risk (5-high, 1-low)	Action (e.g. what action needs to be taken to lower the risk)
1	Will this product contain any new technology with unproven reliability record?	Yes	3 (Moderate)	Implement design optimization, prototyping, real-world testing to evaluate performance, quality control to reduce defects, continuous monitoring, and feedback to refine the new technology.
2	Is this a revolutionary design as opposed to evolutionary?	No	2 (Low)	The technology is evolutionary and poses less risk in the design improvement.
3	Will this design be significantly different from the old one (e.g. more than 30% of the content is new)?	Yes	4 (High)	Need more detailed planning for manufacturing the product and conducting design iterations for further improvements, as well as checking with regulatory compliance.
4	Does this product have any new requirements (e.g. 15 years life instead of 10 years)?	Yes	4 (High)	Need comprehensive testing, such as thermal fatigue failure, wear testing, and failure mode analysis, to understand this new material's behaviour under different conditions, use of protective coatings, and compliance with safety standards
5	Will this product have new applications?	No	1 (Low)	The improved design focused on a single application that involves low risk.
6	Any new materials used in the design?	Yes	3-4 (Moderate to High)	Verifying the material's performance in real-world conditions through testing, implementing strict inspections to improve quality, performing cost analysis, maintaining a reliable supply chain for new material, and ensuring compatibility with existing brake pads.

Task Sheet 5

Programme Risk Assessment

No.	Question	Answer (Yes/No)	Degree of Risk (5-high, 1-low)	Action (e.g. what action needs to be taken to lower the risk)
7	Any changes in the supply chain?	Yes	3 (Moderate)	Diversify suppliers for raw materials for consistent supply, optimise logistics by tracking inventory in real-time, and implement lean manufacturing to reduce delays and shortages.
8	Will the product be made at a different manufacturing location?	Yes	4 (High)	Need to conduct a thorough feasibility study to assess the costs, equipment needs, and supply chain logistics to identify major risks along with work force training.
9	Will this product be supplied to a new customer?	Yes	3 (Moderate)	Prototyping and rigorous testing to gather data and ensure product performance, demonstrate long-term benefits, reliable delivery of the product, and meet customer standards. Provide technical support to assist smooth integration and customer adaptation.
10	Are there any other changes, which can affect reliability?	Yes	3 (Moderate)	Select high-performance brake pads with heat dissipation capabilities, use of high-quality brake fluid, efficient thermal management, and surface coating for corrosion resistance.

Task Sheet 6

HAZOPS



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No.	Name of Unit(s)	Failure/Deviation	Possible Cause(s)	Consequence(s)	Safeguards	Action
1	Disc/ Rotor	No cooling	Air ducts not provided. Wheels provided with steel rims.	Overheating of Rotors	Provide alloy wheels	Add air ducts on the front bumper.
2	Disc/ Rotor	Thermal cracking	Sudden heating and cooling	Reduced braking efficiency, increased stopping distance	New heat dissipation design	Alternative materials with better thermal resistance.
3	Brake caliper	Uneven pad wear	Brake piston sticking	One sided braking, tire wears faster on one side	Regular maintenance	Improve lubrication mechanisms.
4	Brake Pads	Premature wear	Not the right brake pad for the driving condition	Regular replacement, increased maintenance costs	Material and terrain specified on package	Develop brake for all terrain driving conditions.
5	Disc/ Rotor and Brake Pads	Excessive wear	Abrasive particles between pad and rotor, improper pad material	Reduced rotor thickness, risk of fracture	Provide fenders to resist water and dust.	Improve materials of Disc/Rotor and brake pads.
6	Disc/ Rotor	Corrosion	Environmental exposure, salt, moisture	Accelerated wear, uneven pressure on pads causing vehicle to pull to one side	Protective coatings	Develop corrosion-resistant alloys, improve surface treatments
7	Brake caliper	Brake fluid leakage	Seal deterioration	Reduced hydraulic pressure, brake failure	Redundant hydraulic systems	Implement more durable seals, protective shields for caliper components
8	Shim	Noise (squealing)	Vibration between pad and rotor	Brake pad wear, Driver discomfort	Regular washing of car/brake pad regions	Implement improved damping features, redesign shim surface patterns
9	Disc/ Rotor	Voids/inclusions	Casting defects	Structural weakness, crack initiation sites	Quality control	Enhance casting processes, implement quality inspection methods
10	Disc/ Rotor	Warped front rotors	Corrosion/ Excessive heat leaving pad impressions	Steering wheel shake and judder at high speeds	Rotor resurfacing or replacement	Improve rotor material, brake pad material

Task Sheet 6

HAZOPS

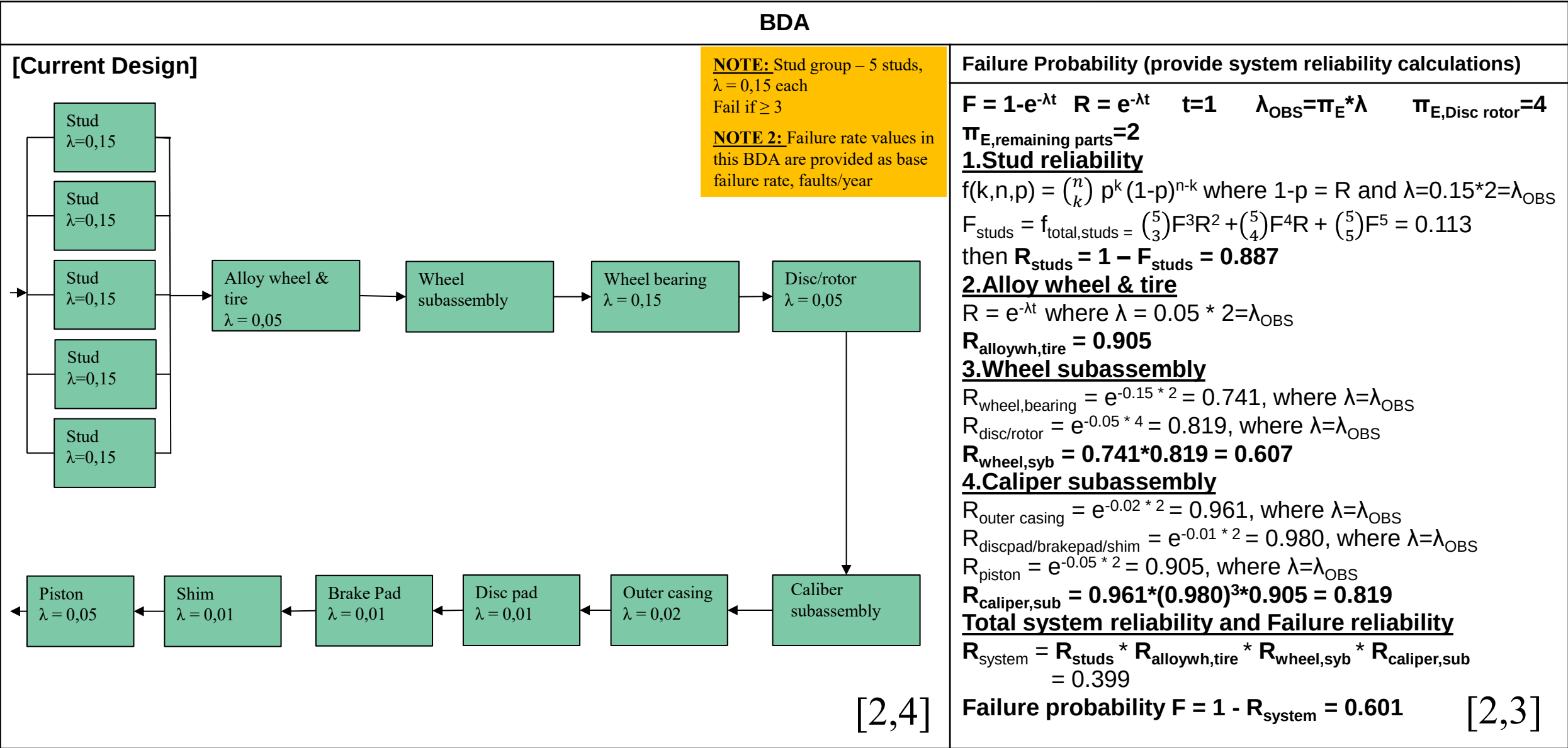


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No.	Name of Unit(s)	Failure/Deviation	Possible Cause(s)	Consequence(s)	Safeguards	Action
11	Brake Fluid	Leak in the hydraulic system	Worn out hydraulic pipes	Loss of hydraulic pressure, brake failure	Regular inspection and maintenance	Using high quality hydraulic systems and connecting parts
12	Disc/ Rotor and brake pads	Increased braking distance	Overloading the vehicle	Brake failure/ Brake pad wear	Keep vehicle loaded within the manual recommended weight.	Instruct driver on the weight limitations/ Display the weight limit on the window stickers
13	Calipers, Master cylinder, brake line(s)	Loss of brake pressure	Naturally worn out/ Manufacturing defects	Uneven braking/ brake failure/spongy brake pedal	Regular inspection and maintenance	Quality control and inspection improvements/ material improvements
14	Caliper assembly	Sudden braking	ABS electronic system failure/ burnt relay	Vehicle skid causing it to topple	Check ABS light regularly	Awareness beeps if ABS light doesn't go off even after driving for few minutes.
15	Caliper assembly	Brake fade	Overheating exceeding dissipation capacity	Reduced braking efficiency, brake pad wear	Current cooling design	Redesigning the ventilation channels, improved cooling solutions
16	Studs	Brake failure/ Wheel disassembly while driving	Improper installation/ stud breakage	Accidents due to brake failure	Proper installation methods, provide marks where the tightened studs should be	Improved material for studs, increasing the number of studs, strengthen the treads on studs.

Task Sheet 7a

BDA



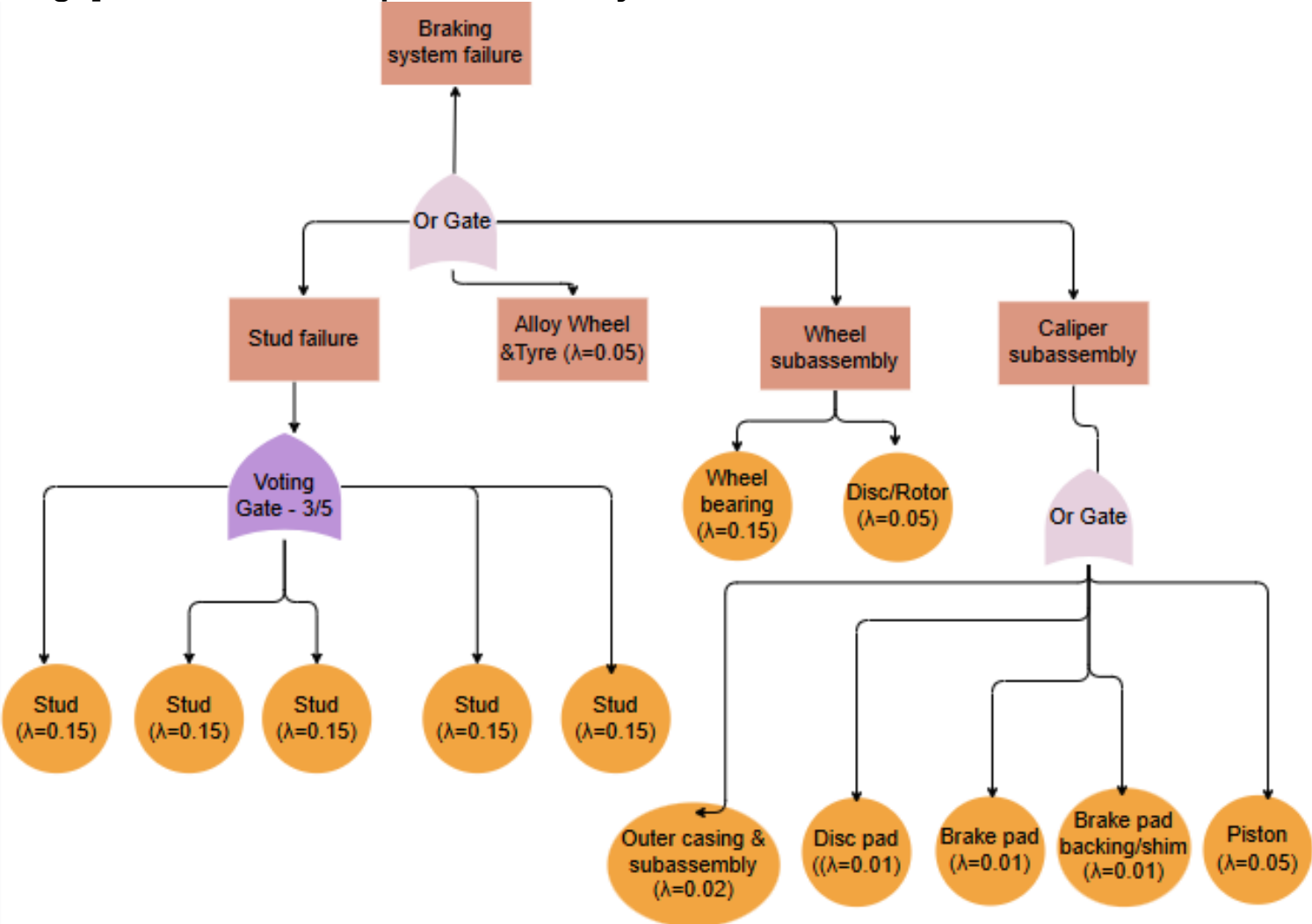
Task Sheet 7a

FTA

FTA

[Current Design] Note: consider top level event ‘System will not function or start’

NOTE : Failure rate values in this FTA are provided as base failure rate, faults/year



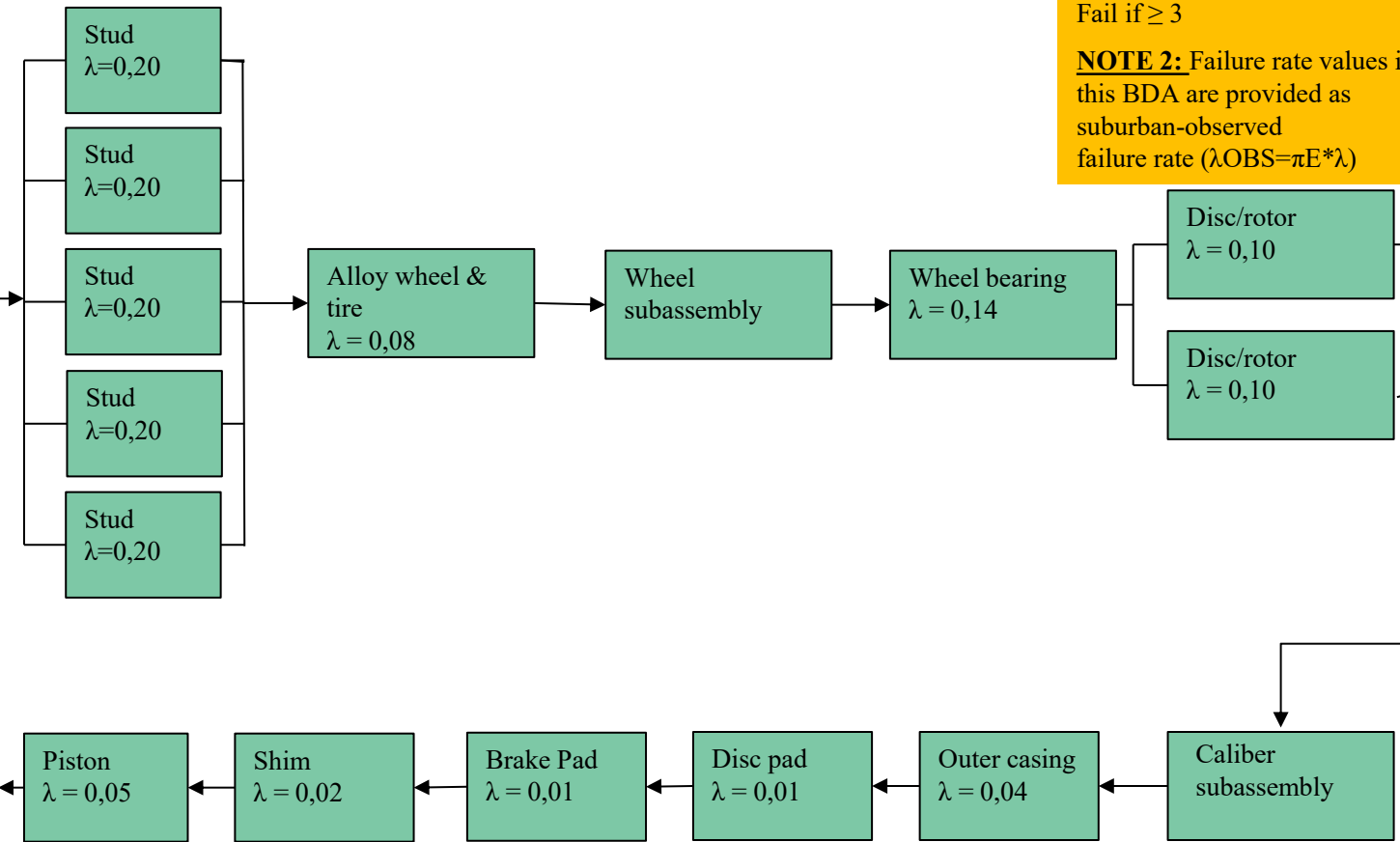
Task Sheet 7b

BDA



BDA

[Improved Design]



NOTE: Stud group – 5 studs, $\lambda = 0,20$ each
Fail if ≥ 3

NOTE 2: Failure rate values in this BDA are provided as suburban-observed failure rate ($\lambda_{OBS} = \pi_E * \lambda$)

Failure Probability (provide system reliability calculations)

$F = 1 - e^{-\lambda t}$ $R = e^{-\lambda t}$ $t=1$ $\lambda_{OBS} = \pi_E * \lambda$ $\pi_{E, Disc \text{ rotor}} = 4$
 $\pi_{E, remaining \text{ parts}} = 2$ All λ used are suburban – observed failure rate (improvement included in the value)

1. Stud reliability

$f(k, n, p) = \binom{n}{k} p^k (1-p)^{n-k}$ where $1-p = R$ and $\lambda = 0.20$

$F_{studs} = f_{total, studs} = \binom{5}{3} F^3 R^2 + \binom{5}{4} F^4 R + \binom{5}{5} F^5 = 0.0444$

then $R_{studs} = 1 - F_{studs} = 0.956$

2. Alloy wheel & tire

$R = e^{-\lambda t}$ where $\lambda = 0.08$

$R_{alloywh, tire} = 0.923$

3. Wheel subassembly

$R_{wheel, bearing} = e^{-0.14} = 0.869$, where $\lambda = 0.14$

$R_{disc/rotor}$ (dual segment), $\lambda = 0.10$ thus $R_{segment} = 0.905$

Then, $R_{disc/rotor} = 1 - (1 - 0.905)^2 = 0.991$ (parallel redundancy)

$R_{wheel, syb} = 0.869 * 0.991 = 0.861$

4. Caliper subassembly

$R_{outer \text{ casing}} = e^{-0.04} = 0.961$, where $\lambda = 0.04$

$R_{discpad} = e^{-0.01} = 0.990$, where $\lambda = 0.01$

$R_{brakepad} = e^{-0.01} = 0.990$, where $\lambda = 0.01$

$R_{shim} = e^{-0.02} = 0.980$, where $\lambda = 0.02$

$R_{piston} = e^{-0.05} = 0.951$, where $\lambda = 0.05$

$R_{caliper, sub} = 0.961 * (0.990)^2 * 0.980 * 0.951 = 0.878$

Total system reliability and Failure reliability

$R_{system} = R_{studs} * R_{alloywh, tire} * R_{wheel, syb} * R_{caliper, sub}$
 ≈ 0.7

Failure probability $F = 1 - R_{system} = 0.3$

[2,4]

[2,3]

Task Sheet 7b

FTA

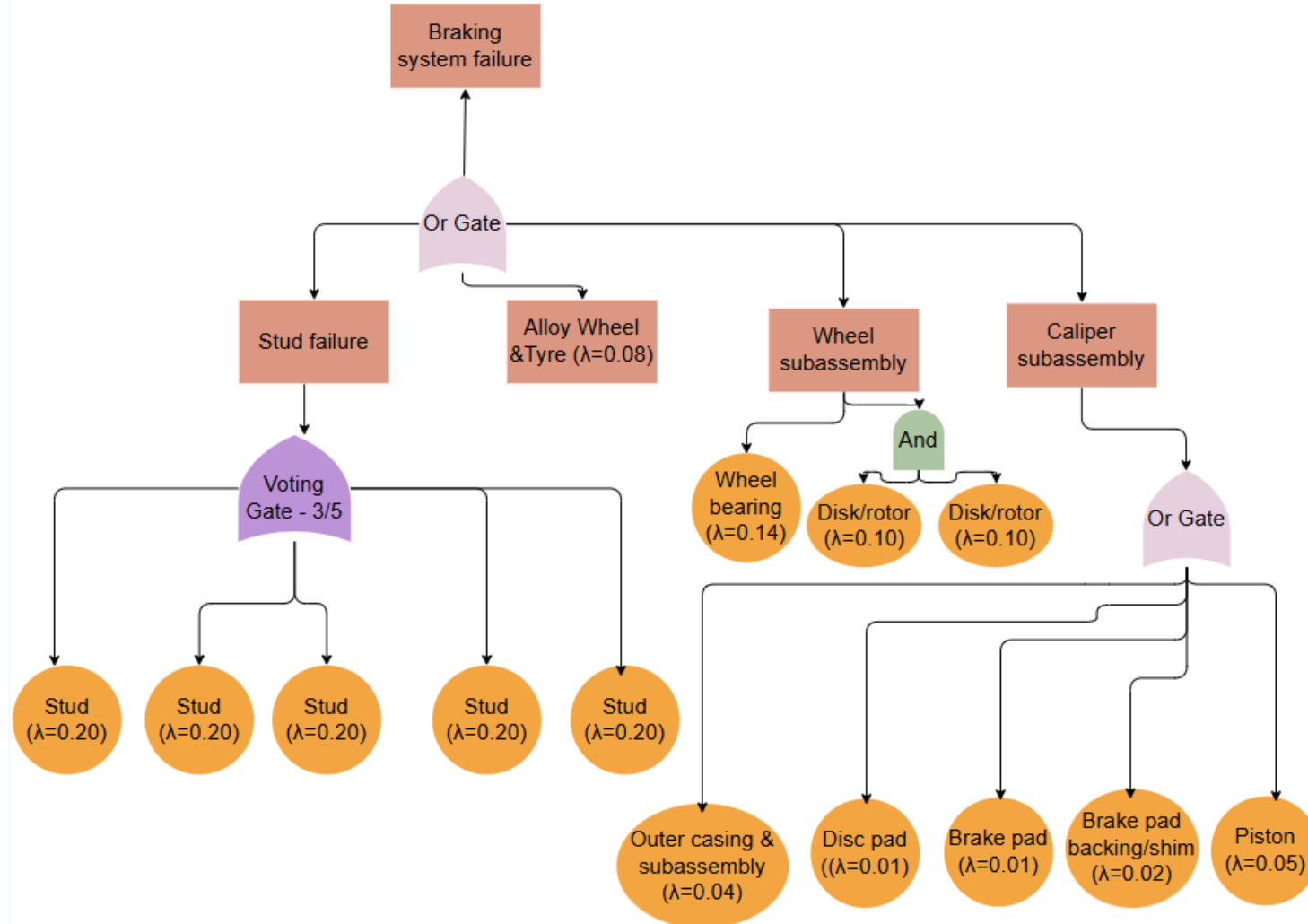


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FTA

[Improved Design] Note: consider top level event 'System will not function or start'

NOTE 1: Failure rate values in this FTA are provided as suburban-observed failure rate ($\lambda_{OBS} = \pi E * \lambda$)



[1,4]

References for Task Sheet 7



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Task Sheet 8

FMEA

Process Step/ Function* Requirements	Potential Failure Mode	Potential Effect(s) of Failure	Severity	Classification	Potential Cause(s) of Failure	Occurrence	Current Process Controls (Prevention)	Current Process Controls (Detection)	Detection	RPN	Recommended Action	Action Results (expected)				
												Actions to be taken & Effective Date	Severity	Occurrence	Detection	RPN
High reliability of braking system	Thermal Cracking	Reduced efficiency in braking, roto/Disc failure due to cracking	9	C r i t i c a l	Excessive heat buildup, poor material selection, worn out brake pads	6	Ventilated/ drilled discs, heat- resistant alloys	Thermal imaging during inspection	5	270	Improving the cooling channels, using higher heat- resistant materials [1]	Improve cooling channels by adding slots or drill holes, use higher heat- resistant materials.	7	5	3	105
High durability in harsh conditions	Warping	Brake judder, uneven braking force	7	M o d e r a t e	Unbalanced wheel alignment, improper torque during installation, potholes on road	5	Proper wheel installation procedures, even torque application on all studs	Torque check during assembly	4	140	Using better alloys with higher thermal stability, ensuring controlled cooling, driving slowly on unknown roads.	Use better alloys with higher strength and thermal stability, ensure controlled cooling, educate on proper installation.	6	4	2	48
Long lifespan and safety	Corrosion	Reduced lifespan, structural weakness	6	M o d e r a t e	Exposure to moisture, still rainwater on roads, poor material coating	6	Protective coatings, stainless steel alloys	Visual inspection for rust and regular maintenance	4	144	Using improved corrosion-resistant coatings, applying anti-rust treatments	Use improved corrosion- resistant coatings, apply anti-rust treatments	5	5	3	75
Consistent performance under load	Excessive Wear	Reduced braking performance, safety risk	8	M a j o r	Frequent high-load braking, poor material selection	7	Wear-resistant alloys, regular inspections	Thickness measurement	6	336	Using particulate ceramic composite brake pads, introducing wear indicators [2]	Use particulate ceramic composite rotors, introduce wear indicators	6	4	2	48
Fail-safe system under stress	Fatigue Failure	Sudden catastrophic failure, complete loss of braking	10	C r i t i c a l	Cyclic loading, material fatigue, improper heat treatment	5	Stress analysis during design, heat treatment quality checks	Visual and ultrasonic inspection	5	250	Improving material properties, using FEA for stress optimization	Improve material properties, use FEA for stress optimization	9	3	2	54

Task Sheet 8

FMEA

Process Step/ Function* Requirements	Potential Failure Mode	Potential Effect(s) of Failure	Severity	Classification	Potential Cause(s) of Failure	Occurrence	Current Process Controls (Prevention)	Current Process Controls (Detection)	Detection	RPN	Recommended Action	Action Results (expected)				
												Actions to be taken & Effective Date	Severity	Occurrence	Detection	RPN
Brake Fluid Selection	Brake Fluid Vaporization	Complete hydraulic failure	9	Critical	Use of hygroscopic fluids and moisture absorption	5	Switch to silicone-based fluid and 2-year fluid replacement [3]	Lab testing of fluid boiling points during manufacturing and conductivity sensors in reservoir	3	135	Install thermal sensors in calipers (Effective: Next service cycle)	Install Arduino-based alarm system and Enforce 2-year fluid replacement (Next service cycle)	8	3	5	120

References

1. Belhocine A, Bouchetara M. Temperature and Thermal Stresses of Vehicles Gray Cast Brake. Journal of Applied Research and Technology. 2013 Oct;11(5):674–82. Available from: [10.1016/S1665-6423\(13\)71575-X](https://doi.org/10.1016/S1665-6423(13)71575-X)
2. Hitesh Bhakuni, Muley AV, Ruchika. Fabrication, testing & analysis of particulate ceramic matrix composite for automotive brake pad application. Materials Today: Proceedings. 2023 Feb 1; Available from: <https://doi.org/10.1016/j.matpr.2023.01.413>
3. Ruff M. Everything you need to know about brake vapour lock [Internet]. Garage Wire. 2020 [cited 2025 Apr 2]. Available from: <https://garagewire.co.uk/news/company/comline/everything-you-need-to-know-about-brake-vapour-lock/>

Task Sheet 9a Load (stress) – Strength Analysis

[Current Design] provide calculations

Material: cast iron

Load/Stress mean ($\bar{x}$) = 180 MPa,
Yield Stress/Strength mean ($\bar{y}$) = 215 MPa,
Standard Deviation (σ_x) = 20 MPa,
Standard Deviation (σ_y) = 20 MPa

Safety Margin (SM):

$$SM = \frac{\bar{y} - \bar{x}}{\sqrt{(\sigma_x^2 + \sigma_y^2)}} = \frac{215 - 180}{\sqrt{20^2 + 20^2}} = \frac{35}{\sqrt{800}}$$

$$SM = 1.237$$

From this SM value, the **Reliability** can be determined by using the **standard cumulative normal distribution function table**:

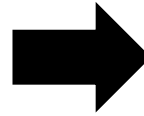
$$\Phi(1.237) = 0.8907$$

Loading Roughness (LR):

$$LR = \frac{\sigma_x}{\sqrt{(\sigma_x^2 + \sigma_y^2)}} = \frac{20}{\sqrt{20^2 + 20^2}} = \frac{20}{\sqrt{800}}$$

$$LR = 0.707$$

Improvement



[Improved Design] provide calculations

Material: Al MMC (6092/SiC/25p-T6) [1]

Load/Stress mean ($\bar{x}$) = 180 MPa,
Yield Stress/Strength mean ($\bar{y}$) = 430 MPa, [1]
Standard Deviation (σ_x) = 20 MPa,
Standard Deviation (σ_y) = 50 Mpa (*calculations are in below slide*)

Safety Margin (SM):

$$SM = \frac{\bar{y} - \bar{x}}{\sqrt{(\sigma_x^2 + \sigma_y^2)}} = \frac{430 - 180}{\sqrt{20^2 + 50^2}} = \frac{250}{\sqrt{1300}}$$

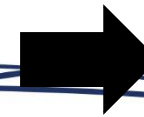
$$SM = 4.642$$

Loading Roughness (LR):

$$LR = \frac{\sigma_x}{\sqrt{(\sigma_x^2 + \sigma_y^2)}} = \frac{20}{\sqrt{20^2 + 50^2}} = \frac{20}{\sqrt{1300}}$$

$$LR = 0.371$$

Improvement





Calculations for Standard deviation (σ_y):

Standard Deviation (σ_y) = $\sqrt{\frac{\sum(xi-\bar{y})^2}{N}}$

- Where,
- xi = individual values in the dataset
- $\bar{y}$ = mean (average) of the dataset
- N = number of values in the dataset
- *For the normal distribution of values, we have chosen the yield stress of range Al MMC (6092/SiC/25p-T6) range between 380 MPa and 480 MPa [1]*
- To find the find mean value $\bar{y}$:
- $\bar{y} = \frac{480+470+450+430+410+390+370}{7}$
- $\bar{y} = 428.57 \approx 430$ MPa
- $(\sigma_y) = \sqrt{\frac{\sum(xi-\bar{y})^2}{N}}$
- Putting values in above equation:

- $\sigma_y = \sqrt{\frac{\sum(480-428.57)^2+(470-428.57)^2+(450-428.57)^2+(430-428.57)^2+(410-428.57)^2+(390-428.57)^2+(370-428.57)^2}{7}}$

$$\sigma_y = \sqrt{2500}$$

$$\sigma_y = 50 \text{ MPa}$$

So, the standard deviation is 50 MPa and mean value is 430 MPa for the Al MMC (6092/SiC/25p-T6)

References:

1. DWA 6092/SiC/25p-T6 Aluminum SiC Metal-Matrix-Composite (Al MMC), Extruded [Internet]. Matweb.com. 2025 [cited 2025 Apr 3]. Available from: <https://www.matweb.com/search/datasheet.aspx?matguid=7a97a2a865544ac3a56bf7144c8dd151>

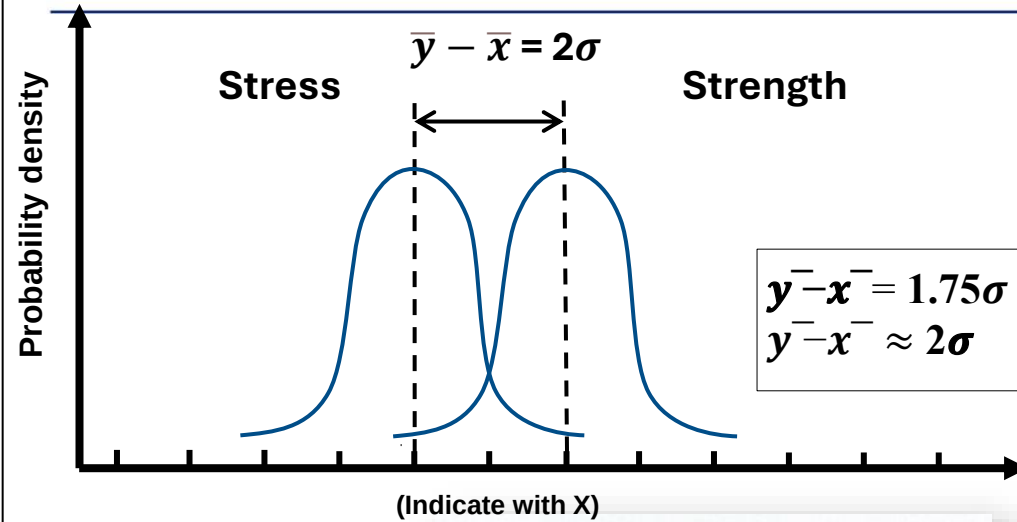
Task Sheet 9b Load (stress) – Strength Analysis



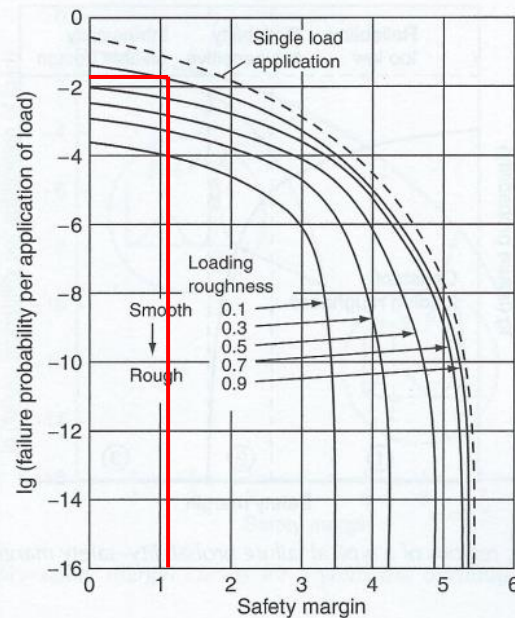
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[Current Design]

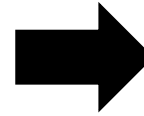
Sketch/illustration of load-strength curves here



- Safety Margin= 1.237
- Loading roughness= 0.707

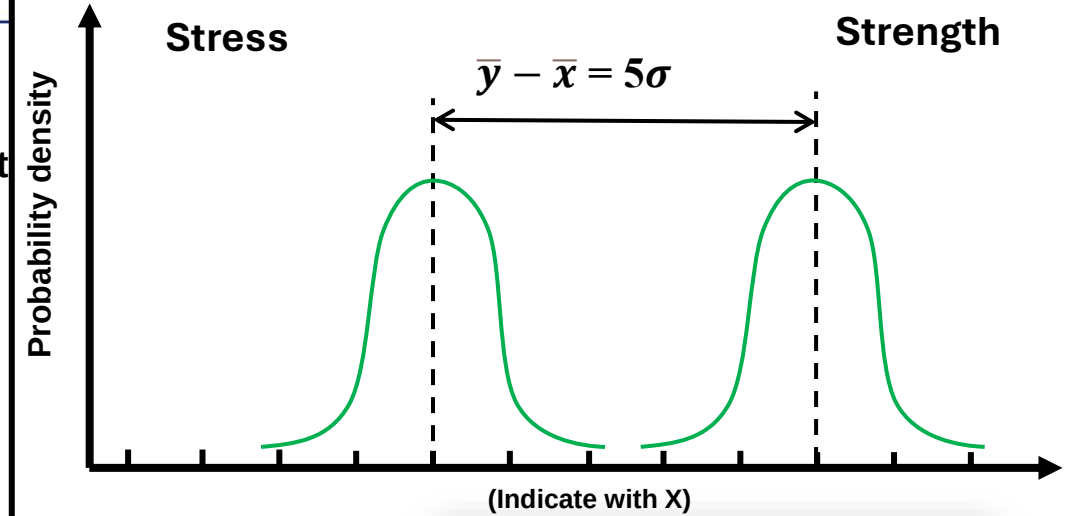


Improvement

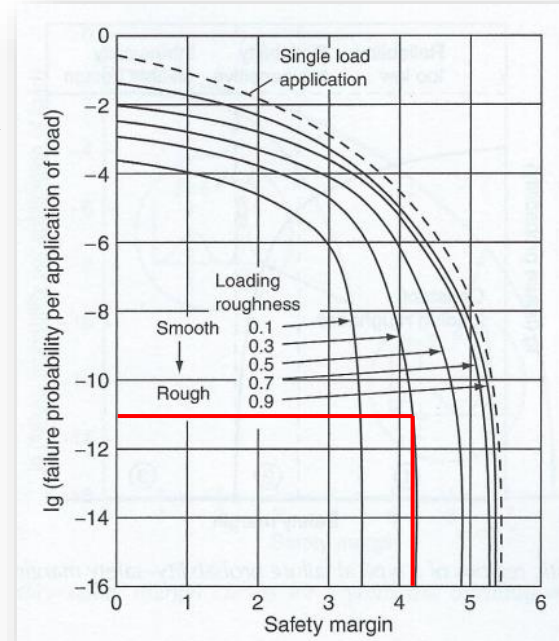


[Improved Design]

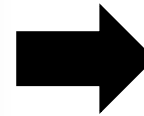
Sketch/illustration of load-strength curves here



- Safety Margin= 4.642
- Loading roughness= 0.371



Improvement

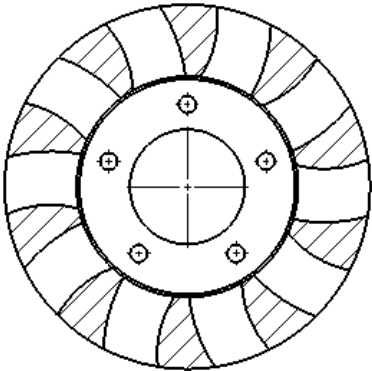
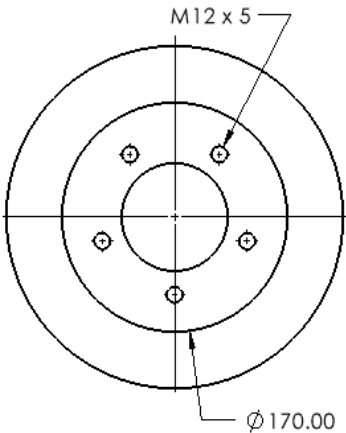
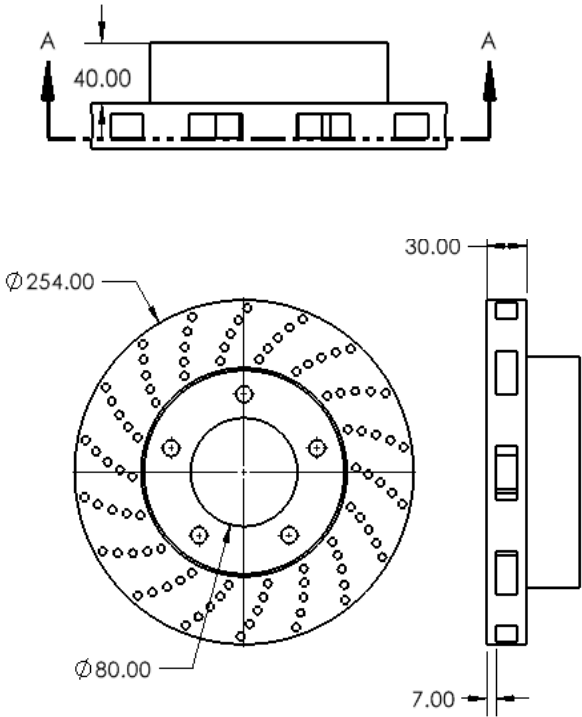


Task Sheet 10 Design

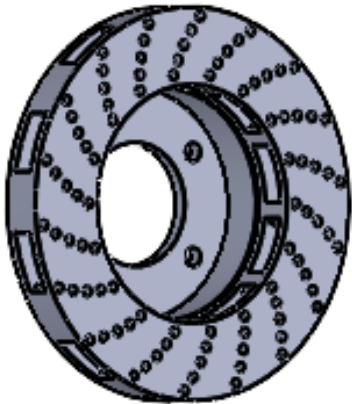
If SolidWorks is used to present the improved design, provide the relevant 2D/3D view images



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SECTION A-A



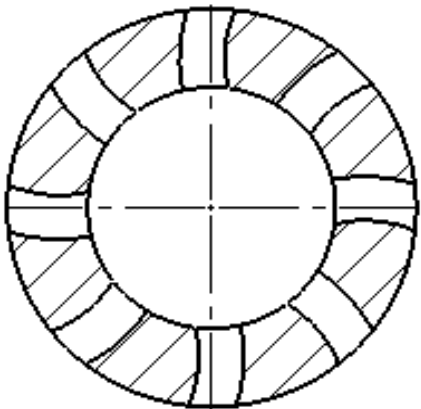
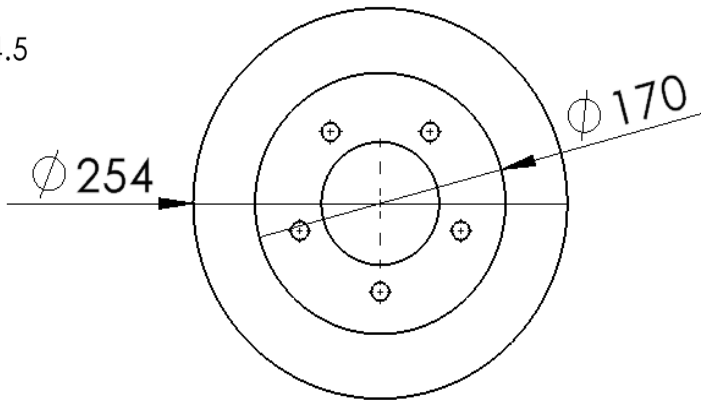
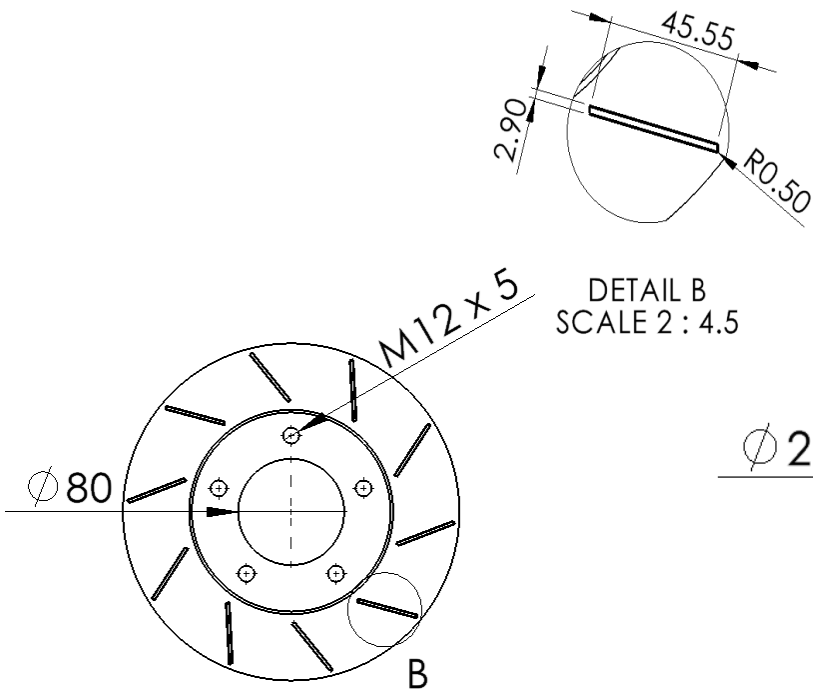
DRAWN		CHECKED		6092/SiC/25p-T6 (Al MMC)				
APPROVED		PROJECTION		MATERIAL DESCRIPTION		COLOUR CODE	SPECIFICATION	NUMBER
				UNLESS OTHERWISE SPECIFIED: ALL DIMENSIONS ARE IN MILLIMETERS		TITLE		
				1 PL. DEC. ±0.2 2 PL. DEC. ±0.01 3 PL. DEC. ±0.005 4 PL. DEC. ±0.0005 UNLESS MIN. & MAX. ARE GIVEN ANGLES ±1° FINISH Ra 6.3 MAX REMOVE BURRS, BREAK SHARP EDGES R 0.2 MAX.		Brake Disc Rotor Design 1		
				WEIGHT:		DRAWING NO. D1		DWG. SIZE A4
						SCALE: N.T.S.		SHEET:

Task Sheet 10 Design

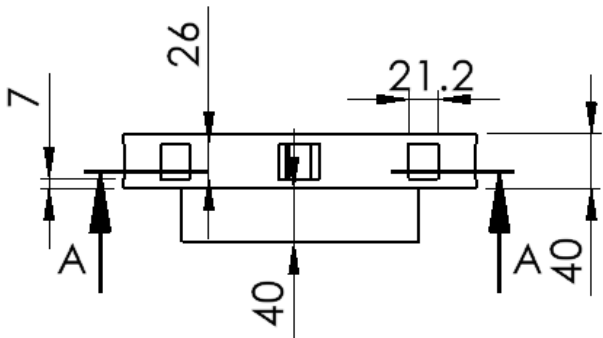
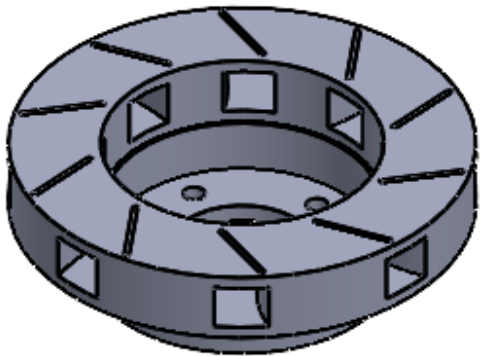
If SolidWorks is used to present the improved design, provide the relevant 2D/3D view images



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SECTION A-A



DRAWN		CHECKED	
APPROVED		PROJECTION	
			
REVISIONS			
SYM	DESCRIPTION		DATE

6092/SiC/25p-T6 (Al MMC)			
MATERIAL DESCRIPTION	COLOUR CODE	SPECIFICATION	NUMBER
UNLESS OTHERWISE SPECIFIED: ALL DIMENSIONS ARE IN MILLIMETERS			
1 PL. DEC. ±0.2 2 PL. DEC. ±0.01 3 PL. DEC. ±0.005 4 PL. DEC. ±0.0005 UNLESS MIN. & MAX. ARE GIVEN ANGLES ±1° FINISH Ra 6.3 MAX REMOVE BURRS, BREAK SHARP EDGES R 0.2 MAX.			
TITLE			DWG. SIZE
Brake Disc Rotor Design 2			A4
DRAWING NO. D2		SCALE: N.T.S.	
WEIGHT:		SHEET:	

Task Sheet 10 Design



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In the process of design improvement, we analyzed different materials based on their composition and mechanical properties. After the benchmarking, we decided to select Aluminum Metal Matrix Composite (AlMMC-6092/SiC/25p-T6) material due to its beneficial properties like corrosion resistance, light weight, and high strength [1]. Along with this, this material can withstand harsh environmental conditions such as high temperature and extreme cold weather without cracking or wear and tear.

The stress-strength analysis carried out on the current and improved design showed that the safety margin for the current design increased significantly due to the higher yield strength of the Al-MMC material. Since the safety margin increased, the area of overlap between the stress and strength curve becomes negligible. This indicates that the probability of failure has reduced dramatically, making the improved design more reliable.

For further improvement, we have made the two drawings of the improved design with different structures, one is a drilled disc rotor, and the other is a slotted disc rotor. In drilled rotors, holes are plotted on the surface of the disc to provide a path for better heat dissipation. During the braking cycle, these holes give passage to the gases built up in the system. This improved heat dissipation reduces brake fading and brake pad glazing. For the slotted disc, slots are made on the rotor surface to evacuate the built-up gas as well as to remove any water that comes in contact with the disc. Another advantage of this type of design is that it ejects brake pad dust easily and avoids the pad glazing, which ultimately provides better braking performance. [2] Overall, the rotor disc with a slotted pattern, made up of higher strength material (Al-MMC) and a greater safety margin, is relatively better than most of the competitors mentioned in the benchmarks.

References:



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1. DWA 6092/SiC/25p-T6 Aluminum SiC Metal-Matrix-Composite (Al MMC), Extruded [Internet]. Matweb.com. 2025 [cited 2025 Apr 3]. Available from: <https://www.matweb.com/search/datasheet.aspx?matguid=7a97a2a865544ac3a56bf7144c8dd151>
 2. Silva SAMD, Kallon DVV. FEA on different disc brake rotors. Procedia Manufacturing. 2019;35:181–6.
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